

Digital Certificate

This week's project we will be creating a digital certificate. Developing a digital certificate that aligns with industry standard processes and employs robust encryption keys is essential for establishing trust and security in online communications. OpenSSL, a versatile and widely used toolkit, provides a reliable framework for creating these certificates on the Windows platform. By following a series of systematic steps within OpenSSL, users can generate a secure digital certificate that not only adheres to recognized standards but also ensures the confidentiality and integrity of sensitive data exchanged over the digital landscape. I will be providing a walk-through guide on how to create a digital certificate using Openssl on Windows.

Using Openssl in command prompt as administrator type in this command:

```
openssl req -x509 -newkey rsa:2048 -keyout key.pem -out cert.pem -nodes -days 365
```

req x509 for the certificate request, newkey to establish that it's a new key for the certificate using rsa 2048 bit, and keyout which is going to be the key file in which they get stored.

[illegible]

This will prompt some questions for the digital certificate.

Simply enter your country name, state, organization, unit name, common name, and email address.

```
Country Name (2 letter code) [AU]:US
State or Province Name (full name) [Some-State]:Oklahoma
Locality Name (eg, city) []:Norman
Organization Name (eg, company) [Internet Widgits Pty Ltd]:OSU-IT
Organizational Unit Name (eg, section) []:Cryptography
Common Name (e.g. server FQDN or YOUR name) []:Jordan
Email Address []:iflaum@okstate.edu
```

Next, we need to put them into a single file using the command:

```
openssl pkcs12 -export -in cert.pem -inkey key.pem -out test-  
cert.pfx.
```

```
C:\Windows\System32>openssl pkcs12 -export -in cert.pem -inkey key.pem -out test-cert.pfx
Enter Export Password:
Verifying - Enter Export Password:
```

It's going to ask you for a password simply type in a password you will remember and verify it again.

Export a public key by using the command:

public.pem

Also, use the command to view the digital certificate you've created:

More test-cert-public.pem

```
C:\Windows\System32>>openssl pkcs12 -in test-cert.pfx -clcerts -nokeys -out test-cert-public.pem
Enter Import Password:

C:\Windows\System32>more test-cert-public.pem
Bag Attributes
    localKeyID: D5 88 28 B4 5E D9 12 30 AB 19 03 61 34 F5 24 41 33 58 39 55
subject=C = US, ST = Oklahoma, L = Norman, O = OSU-IT, OU = Cryptography, CN = Jordan, emailAddress = jflaum@okstate.edu
issuer=C = US, ST = Oklahoma, L = Norman, O = OSU-IT, OU = Cryptography, CN = Jordan, emailAddress = jflaum@okstate.edu
-----BEGIN CERTIFICATE-----
MIID/TCCAuIwAgwIBAgIUDBsEQIEfgy24QDvFnjR6mAYStqAwDQYJKoZIhvcNAQEL
BQAwY0xkCzABAgNBAYTA1VMTREwDwYDVQQIDAhPa2xhaG9tYTEPMA0GA1UEBmwwG
Tm9ybiUwMQ8wDQYDVQQKDAZPU1UtSV0xFTATBgNVBAsMDENyeXB0b2dyYXBoeTEP
MA0GA1UEAwSm93Y2ZGUwSEwHYK0ZThvcNAQkBFHj3Qmxhdw1Ab2t2ZGF0ZS51
ZHUwHhcnMjMwODExMTYxODQ0W4hcnMjMwODEwMTYxODQ0W4jCBjTELMAKGA1UEBHMCM
VVMxETEAPBgNVBAGMCE9rbGob2t1hMQ8wDQYDVQQHDAZOb3J3Y2xkZANBGA1UEAOM
Bk9TYW91MjVDEwVBMGA1UECwwMQ3J3cHRvZ3JhcGhSMHQ8wDQYDVQQDDAZKb3J3Y2xk
ITAFBgkqhkiG9w0BCQEWEmpmbGF1bU9va3N0YXR1LWkVkdTCCASTwDQYJKoZIhvcN
AQEBBQADAggEPADCCAQoCggEBANnPeB6jtNxiu1stJXCEZByDVhzLNUGMYf8HI2L
qp2cKKhl8Q-J1ePsbZgfGLr4966YgqQvg17a+mQX9rmYBCKWw6q1GET/cvtmLtiF
suuxUPoAtB4B67NueVMcye3PhqajGe0yFjIjCRRxFEM1eZph8i3PXS Mk2XF5QWAWQ
Ead79RuZ225gDxPefus6/iLvJw+TySKgBBpWw5o0+JgBK0QkKbsw48Uq2Ypvfth
5V32dfp4Sbae84sJZ5UU8Ed5h60dMRESDozdTMeLxZD0meu7oJvuBno13XSXSpqZI
wZFN+Cqm0KM001qJW108wRmCNxg44U989HPVe1n0Jc69DECAwEAANtMFEwHQYD
VR0oBmFEBjMjDF+jCgFIYhUKnHDTUjtFdUckMB8GA1UdIwQYMBaAFBjMjDF+jCgFI
YhUKnHDTUjtFdUckMA8GA1UdEwEB/wQFMAMBAF8wDQYJKoZIhvcNAQELBQADggEB
AiwKN2S7D1NpysjPE1Cx++2Ui9yHtdTumaWVLZkxx2gY2XXdc5NF0w/EVYUnHF1
JxogI1vfnFy2KVRkzF6iaJZxs5EZCkuxIti1vq/imEixjFLnk1yGwXZzxsjw
MASY446oH1dF5Yy6B8Buts84pgM7uygwqooZBrKA2BYkNsG21B2BZV4VMi1R4TM8E
jgcBcXqAF7dP6nbDiUnG7HugC7utydvho03oboOwdoSf5bZthZJQb0zAY80tTS
6jkGjklpKNubV+9hrazFuuiCSUwiJwChPwdrTrPJc/fgGBTH3VUCW3x+sGpf8QyP
J1YdmooDNcmPjD5QpQe6Bk=
-----END CERTIFICATE-----
```

The file test-cert.pfx has our private key and our certificate:

More test-cert.pfx

[illegible]

Import wizard:



  Certificate Import Wizard

Welcome to the Certificate Import Wizard

This wizard helps you copy certificates, certificate trust lists, and certificate revocation lists from your disk to a certificate store.

A certificate, which is issued by a certification authority, is a confirmation of your identity and contains information used to protect data or to establish secure network connections. A certificate store is the system area where certificates are kept.

Store Location

☒ Current User

☐ Local Machine

To continue, click Next.

Next

Cancel

Private key protection

To maintain security, the private key was protected with a password.

Type the password for the private key.

Password:

••••

☐ Display Password

Import options:

- ☐ Enable strong private key protection. You will be prompted every time the private key is used by an application if you enable this option.
- ☐ Mark this key as exportable. This will allow you to back up or transport your keys at a later time.
- ☐ Protect private key using virtualized-based security(Non-exportable)
- ☒ Include all extended properties.

Next

Cancel



←  Certificate Import Wizard

File to Import

Specify the file you want to import.

File name:

Note: More than one certificate can be stored in a single file in the following formats:

Personal Information Exchange- PKCS #12 (.PFX,.P12)

Cryptographic Message Syntax Standard- PKCS #7 Certificates (.P7B)

Microsoft Serialized Certificate Store (.SST)

Completing the Certificate Import Wizard

The certificate will be imported after you click Finish.

You have specified the following settings:

Certificate Store Selected	Automatically determined by the wizard
Content	PFX
File Name	C:\Users\Chron\OneDrive\Desktop\test-cert.pfx

Finish

Cancel

|-----BEGIN ·PRIVATE ·KEY-----¶
MIIEvAIBADANBgkqhkiG9w0BAQEFAASCbKYwggSiAgEAAoIBAQDZz3geo7Tcbopb¶
krSVvhGQcglYcyzVBjGH/B4ti6qdnCpIS/EPiZXj7G2YHxi6+PeumIKkL4Ne2vpk¶
F/a5mAXCIsOqtRhLf3L7Zi7YhbLsMVD6ALQeAeu5lHlTHMntz4amoxntMnyYiXEc¶
XxDJXmaYfItz10jJNlxeUFgFkBGne/UVm9tuYA8Tz3n7rOv4i741vk8kpIAQaVsO¶
adviYAZEDpCm7FuPFKtmKb37YeVd9nX6eEm2nvOLCWeVFPBHeYetHTEREg6M3UzH¶
i8WQzpnru6Cb7gZ6Nd0l0qamSMGXzfgqpltCjNKNaiVtdPMEZgjcYOOFPfPRz1Xp¶
Z9CXOvQxAgMBAAECgf8z1qz25v+ACWFSRYX+Z7HrCl6Qd7RqtbM4vG0EjpsMydT9¶
O17tQXY0SNoVtem/LwINC9F5r9U49Qmi7X7dY/Wx2lSSzGJ73qKRV2UX/roWxe8k¶
bJi6lLozbd5NssM36IfiIS34M9UF5ouptPaArXcGry/vKzuC8TZTz1229TeHklGV¶
poC85p/ZwhsA6A5LmotNssimHIPv8FQNqPL/Pg6XMvtsCypKlGUuLASwVmV0iJwy¶
XuiLda2h+334pvzEuCyFPW2Wh3J6KlpfsGYymcCNOrfMmt0nHNc3SXhHBRbNx2AU¶
OBoz4qulvZkdTzHjRi3UMrhTjedlQFkvSeCXGECgYEA98c7i9UdyWj8GD47XPRG¶
UIPPhCxiDYwju+wb3bvvggMax/nIFkuiFXxY7CbWV+ZuiHqQfgFtJH4Zdd7XSNo1¶
kYQWOaTGin9hXlghHUOz1crRUD/xvGfRt/GKIDo65lxB2ypL+uTAjeqjRRDlBXyz¶
OjOzS0WwEbxgWmplBVqGXnkCgYEA4QmsUF49ctQif0CLa1rvXnqXsTOF4kGJ1TAn¶
JHfdlRYo5i3Iqxiz09htPw7GBJWJdC9BFsrIp/lHp3o2Bkv74CTjKypy+8XiFJX0¶
dvUrQiLeYHIJtaGlTFbcTW3l5scOgwncKl+mYs9Efogq/TVL4u8fiJl/XfF5XLPb¶
N8ojdXkCgYEAqYc/TqU6BBw4n720kBzyHxSQXzaev/YqXOFnD4cOkD3fjQNxyw9k¶
jp7NmIXaEbovlmFFi3EeY+GDOgqF8eqazD07l6UN0qTk0n9R6Nvxp01leSrHEslV¶
zC00/I3aztJAnNuLsNXiiTcJDrBIgkCXwDKvccj0dq4Unsg85jTRysECgYATUz1U¶
+YLVZZpAy04tvfVoa6azNPV+ZJptkGEMJK8MKYcBnVTqPIjU3HU1QSOIKT9Qlu5E¶
As4QfrvEih4xlDzhFsnbXx3YuTmdAvOnc45vc4cdSS3ER0lmsN7iVlmy480/O42I¶
pgYn3hzHsPHgJL+LkXVe/nonmyEM8PUWaw16IQKBgQCTUxCikOF+x228rJgTjVhQ¶
3Zdj9aq7aMvJozpBk3dMMfyM2ZybTPIRV8VqEcZl46l3ybdP4fueGi0PezIkaLM9¶
MZYFWWsmsfwMxWx1VVzmzD9n3KEXZgODxf6vbaQDPDwbFQCZLERhZtneAvjzWJ6c¶
ywF/5qkzXvpLR+WaV222XA==¶
-----END ·PRIVATE ·KEY-----¶

-----BEGIN CERTIFICATE-----
MIID/TCCAuwgAwIBAgIUDBsEQiEfgy24qDvFnjR6mAY5tqAwDQYJKoZIhvcNAQELQg
BQAwgY0xCzAJBgNVBAYTAlVTMREwDwYDVQQIDAhPa2xhaG9tYTEPMA0GA1UEBwwGQg
Tm9ybWFuMQ8wDQYDVQQKDAZPU1UtSVQxFTATBgNVBAsMDENyeXB0b2dyYXBoeTEPQg
MA0GA1UEAwwGSm9yZGFuMSEwHwYJKoZIhvcNAQkBFhJqZmxhdW1Ab2t2dGF0ZS51Qg
ZHUwHhcNMjMwODExMTYxODQ4WhcNMjQwODEwMTYxODQ4WjCBjTELMAkGA1UEBhMCQg
VVMxETAPBgNVBAGMCE9rbGFob21hMQ8wDQYDVQQHDAZOb3JtYW4xDzANBgNVBAoM
Bk9TVS1JVDEVMBMGA1UECwwMQ3J5cHRvZ3JhcGh5MQ8wDQYDVQQDDAZKb3JkYW4xQg
ITAfBgkqhkiG9w0BCQEWEmpmbGF1bUBva3N0YXRlLmVkdTCCASIwDQYJKoZIhvcN
AQEBBQADggEPADCCAQoCggEBANnPeB6jtNxuiluStJXCEZByDVhzLNUGMYf8Hi2LQg
qp2cKkhL8Q+JlePsbZgfGLr4966YgqQvg17a+mQX9rmYBcKWw6q1GEt/cvtmLtiFQg
suwxUPoAtB4B67nUeVMcye3PhqajGe0yFJiJcRxfEMleZph8i3PXSMk2XF5QWAWQg
Ead79RUz225gDxPPefus6/iLvJW+TySkgBBpWw5oO+JgBkQOkKbsW48Uq2YpvfthQg
5V32dfp4Sbae84sJZ5UU8Ed5h60dMRESDozdTMeLxZDomeu7oJvuBno13SXSpgZIQg
wZfN+CqmW0KM0o1qJW108wRmCNxg44U989HPVelnoJc69DECAwEAAaNTMFEwHQYDQg
VR0OBBYEFBMJdf+jCgfIYhUKnHDTUjTfDuckMB8GA1UdIwQYMBaAFBMJdf+jCgfIYh
YhUKnHDTUjTfDuckMA8GA1UdEwEB/wQFMAMBAf8wDQYJKoZIhvcNAQELBQADggEBQg
AIWnKN2S7D1YNpsjPElCx++2Ui9yHtdTUmaWVLZkxx2gY2XXdc5NF0w/EVYUnHF1Qg
JxogIplvnfFnFy2KVRkzF6IaJZXs5EZCkuxItilvq/imEIXjFLnklyGWXUZzxsjwQg
WAsY4/6oHldF5YY6BButs84ppMUTgwqooXZBrKA2BykNsG21B2BZV4VMiLR4tM8EQg
jgcbCx5oqAF7dP6nbDiUnG7HUGC7uyDvh0n3oboOwdoSfsbZthzJQb0zAuY80tTSQg
6jkGjkplKNubV+9hrzFuuiCSUWijWcHpwdTrPJC/fgGBTH3yVUCW3x+sGpf8QyPQg
JlYBdmOdnCmPjD5qPqe6Bk=Qg
-----END CERTIFICATE-----

```

Bag.Attributes¶
...localKeyID: D5 88 28 B4 5E D9 12 30 AB 19 03 61 34 F5 24 41 33 58 39 55 ¶
subject=C := US, ST := Oklahoma, L := Norman, O := OSU-IT, OU := Cryptography, CN := Jordan, emailAddress := jflaum@okstate.edu¶
issuer=C := US, ST := Oklahoma, L := Norman, O := OSU-IT, OU := Cryptography, CN := Jordan, emailAddress := jflaum@okstate.edu¶
-----BEGIN CERTIFICATE-----¶
MIID/TCCAuWgAwIBAgIUDBsEQiEfgy24qDvFnjR6mAY5tqAwDQYJKoZIhvcNAQEL¶
BQAwgY0xCzAJBgNVBAYTAlVTMREwDwYDVQQIDAhPa2xhaG9tYTEPMA0GA1UEBwwG¶
Tm9ybWFuMQ8wDQYDVQQKDAZPU1UtSVQxFTATBgNVBAsMDENyeXB0b2dyYXB0eTEP¶
MA0GA1UEAwGSm9yZGFuMSEwHwYJKoZIhvcNAQkBFhJqZmxhdW1Ab2tzdGF0ZS51¶
ZHUwHhcNMjMwODExMTYxODQ4WhcNMjMwODEwMTYxODQ4WjCBjTELMAkGA1UEBhMC¶
VVMxETAPBgNVBAGMCE9rbGFob21hMQ8wDQYDVQQHDAZOb3JtYW4xDzANBgNVBAoM¶
Bk9TVS1JVDEVMBMGA1UECwwMQ3J5cHRvZ3JhcGh5MQ8wDQYDVQQDDAZKb3JkYW4x¶
ITAFBgkqhkiG9w0BCQEWEmpmbGF1bUBva3N0YXRlLmVkdTCCASIwDQYJKoZIhvcN¶
AQEBBQADggEPADCCAQoCggEBANnPeB6jtNxuiluStJXCEZByDVhzLNUGMYf8Hi2L¶
qp2cKkhL8Q+JlePsbZgfgLr4966YgqQvg17a+mQX9rmYBcKWw6q1GET/cvtmLtiF¶
suwxUPoAtB4B67nUeVMcye3PhqajGe0yfJiJcRxfEMleZph8i3PXSMk2XF5QWAWQ¶
Ead79RUz225gDxPPefus6/iLvJw+TySkgBBpWw5oO+JgBkQOkKbsW48Uq2Ypvfth¶
5V32dfp4Sbae84sJZ5UU8Ed5h60dMRESDozdTMeLxZDOmeu7oJvuBno13SXSpq2I¶
wZfN+CqmW0KM0o1qJW108wRmCNxg44U989HPVelnoJc69DECAwEAAaNTMFEwHQYD¶
VR0OBBYEfEBMJdf+jCgfIYhUKnHDTUjTfDuckMB8GA1UdIwQYMBaAFBMJdf+jCgfI¶
YhUKnHDTUjTfDuckMA8GA1UdEwEB/wQFMAMBAf8wDQYJKoZIhvcNAQELBQADggEB¶
AIWnKN2S7D1YNpsjPE1Cx++2Ui9yHtdTUmaWVLZkxx2gY2XXdc5NF0w/EVyUnHF1¶
JxogIplvnfFnFy2KVRkzF6IaJZXs5EZCkuxItilvq/imEIxjFLnklyGWXUZzxsjw¶
WAsY4/6oHldF5YY6BButs84ppMUTgwqooXZBrKA2BykNsG21B2BZV4VMILR4tM8E¶
jgcbCx5oqAF7dP6nbDiUnG7HUGC7uyDvh0n3oboOwdoSfsbZthzJQb0zAuY80tTS¶
6jkGjkplKNubV+9hrzFuuiCSUWijWcHpwdTrPJC/fgGBTH3yVUCW3x+sGpf8QyP¶
JLYBdmoODNcmPjD5qPqe6Bk=¶
-----END CERTIFICATE-----¶

```

SHA-256 Flag Validation

Validating Flag Files

To validate you need to open command prompt and navigate to the folder containing the “FlagFiles”.

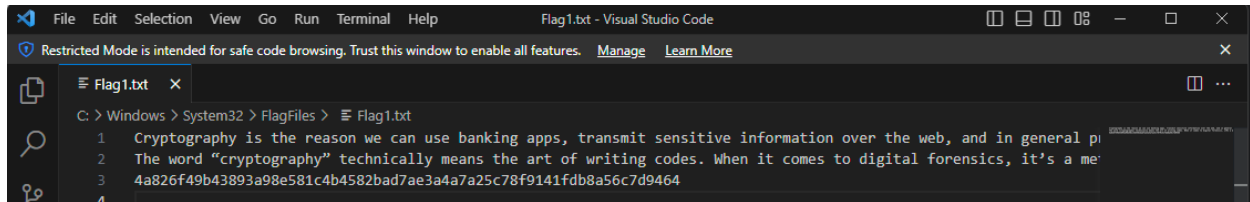
Calculate the SHA-256 hash of the flag file using the command:

```
openssl dgst -sha256 flag_file.txt
```

Correct hash from flag_file.txt

e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855

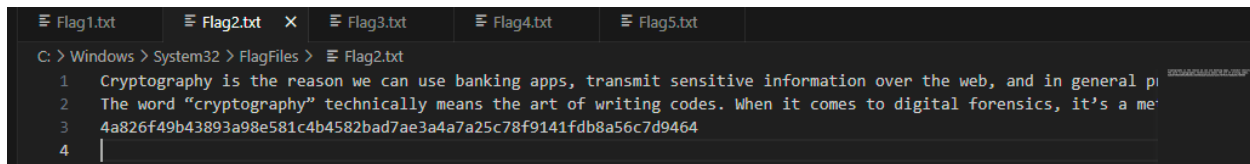
Flag1.txt

A screenshot of the Visual Studio Code editor. The title bar says 'Flag1.txt - Visual Studio Code'. Below the title bar, there's a message: 'Restricted Mode is intended for safe code browsing. Trust this window to enable all features. Manage Learn More'. The editor shows a file named 'Flag1.txt' with the following content:

```
C: > Windows > System32 > FlagFiles > Flag1.txt
1 Cryptography is the reason we can use banking apps, transmit sensitive information over the web, and in general pi
2 The word "cryptography" technically means the art of writing codes. When it comes to digital forensics, it's a me'
3 4a826f49b43893a98e581c4b4582bad7ae3a4a7a25c78f9141fdb8a56c7d9464
4
```

Result: Invalid

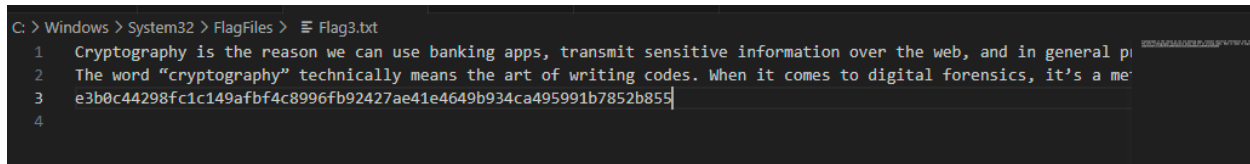
Flag2.txt

A screenshot of the Visual Studio Code editor showing multiple tabs: 'Flag1.txt', 'Flag2.txt' (active), 'Flag3.txt', 'Flag4.txt', and 'Flag5.txt'. The active tab 'Flag2.txt' shows the following content:

```
C: > Windows > System32 > FlagFiles > Flag2.txt
1 Cryptography is the reason we can use banking apps, transmit sensitive information over the web, and in general pi
2 The word "cryptography" technically means the art of writing codes. When it comes to digital forensics, it's a me'
3 4a826f49b43893a98e581c4b4582bad7ae3a4a7a25c78f9141fdb8a56c7d9464
4
```

Result: Invalid

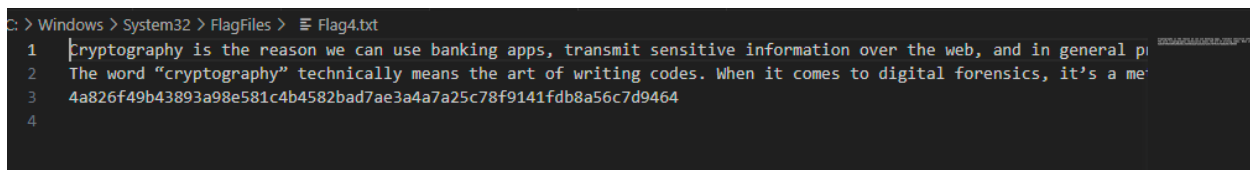
Flag3.txt

A screenshot of the Visual Studio Code editor showing the 'Flag3.txt' tab. The content is as follows:

```
C: > Windows > System32 > FlagFiles > Flag3.txt
1 Cryptography is the reason we can use banking apps, transmit sensitive information over the web, and in general pi
2 The word "cryptography" technically means the art of writing codes. When it comes to digital forensics, it's a me'
3 e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855
4
```

Result: Valid SUCCES this is the True Flag!

Flag4.txt

A screenshot of the Visual Studio Code editor showing the 'Flag4.txt' tab. The content is as follows:

```
C: > Windows > System32 > FlagFiles > Flag4.txt
1 Cryptography is the reason we can use banking apps, transmit sensitive information over the web, and in general pi
2 The word "cryptography" technically means the art of writing codes. When it comes to digital forensics, it's a me'
3 4a826f49b43893a98e581c4b4582bad7ae3a4a7a25c78f9141fdb8a56c7d9464
4
```

Result: Invalid

Flag5.txt

```
C: > Windows > System32 > FlagFiles > Flag5.txt
1  Cryptography is the reason we can use banking apps, transmit sensitive information over the web, and in general p
2  The word "cryptography" technically means the art of writing codes. When it comes to digital forensics, it's a me
3  4a826f49b43893a98e581c4b4582bad7ae3a4a7a25c78f9141fdb8a56c7d9464
4
```

Result: Invalid

This approach ensures that the flag files cannot be distinguished based on visual inspection or file attributes. The validation relies solely on the calculated hash matching the stored validation hash.